Bereich: Strings

Quersumme	**Musterlösung**
Package: `de.dhbwka.java.exercise.strings`	**Klasse:** `CrossTotal`

```java
package de.dhbwka.java.exercise.strings;

import java.util.Scanner;

/**
 * @author DHBW lecturer
 * @version 1.0
 *
 * Part of lectures on 'Programming in Java'.
 * Baden-Wuerttemberg Cooperative State University.
 *
 * (C) 2015 by J. Sidler, T. Schlachter, C. Schmitt, W. Süß
 */
public class CrossTotal {

	public static void main(String[] args) {
		Scanner scan = new Scanner(System.in);
		System.out.print("Bitte Zahl für Quersumme eingeben: ");
		String number = scan.next();
		int total = 0;
		for(char c : number.toCharArray())
			total += Character.getNumericValue(c);
		// Alternative Implementierung
//		for (char c : number.toCharArray())
//			total += (c - '0');
		System.out.println("Die Quersumme von " + number +
			" ist " + total);
		scan.close();
	}
}
```

Bereich: Strings

Palindrom	**Musterlösung**
Package: `de.dhbwka.java.exercise.strings`	**Klasse:** `Palindrome`

```java
package de.dhbwka.java.exercise.strings;

import java.util.Scanner;

/**
 * @author DHBW lecturer
 * @version 1.1
 *
 * Part of lectures on 'Programming in Java'. Baden-Wuerttemberg
 * Cooperative State University.
 *
 * (C) 2015-2024 by J. Sidler, T. Schlachter, C. Schmitt, W. Süß
 */
public class Palindrome {

	public static void main(String[] args) {
		Scanner scan = new Scanner(System.in);
		System.out.print("Bitte Wort eingeben: ");
		String word = scan.nextLine();
		StringBuilder reverse = Palindrome.reverse(word);
		// Alternativ
//		StringBuilder reverse = new StringBuilder(word).reverse(); // RTFM!
		System.out.println("Umgekehrt: " + reverse);
		System.out.println(word + " ist "
		+ (word.equalsIgnoreCase(reverse.toString()) ? "" : "k")
		+ "ein Palindrom.");
		scan.close();
	}

	public static StringBuilder reverse(String s) {
		StringBuilder sb = new StringBuilder();
		for (int i=s.length()-1; i>=0; i--)
			sb.append(s.charAt(i));
		return sb;
	}

}
```

Bereich: Strings

Römische Zahlen	**Musterlösung**
Package: `de.dhbwka.java.exercise.strings`	**Klasse:** `RomanNumber`

```java
package de.dhbwka.java.exercise.strings;

import java.util.Scanner;

/**
 * @author DHBW lecturer
 * @version 1.0
 *
 * Part of lectures on 'Programming in Java'.
 * Baden-Wuerttemberg Cooperative State University.
 *
 * (C) 2015 by J. Sidler, T. Schlachter, C. Schmitt, W. Süß
 */
public class RomanNumber {

    /** Get the decimal value of a Roman digit.
     *  0 is returned for invalid chars.
     *  @param c Roman digit (char)
     *  @return decimal value of a single Roman digit */
    private static int getValue(char c) {
        switch (Character.toUpperCase(c)) {
            case 'I' : return 1;
            case 'V' : return 5;
            case 'X' : return 10;
            case 'L' : return 50;
            case 'C' : return 100;
            case 'D' : return 500;
            case 'M' : return 1000;
        }
        return 0;
    }

    /** Get the decimal value of a Roman number.
     *  The syntax is not checked.
     *  @param c Roman number (String)
     *  @return decimal value of the Roman number */
    public static int getValue(String s) {
        int result = 0;
        // Add all numbers but the last
        for(int i=0; i<s.length()-1; i++) {
            int valueAti = getValue(s.charAt(i));
            result += (valueAti < getValue(s.charAt(i+1)))
                      ? -valueAti :  valueAti;
        }
        // Add last number
        result += getValue(s.charAt(s.length()-1));
        return result;
    }
```

```
    public static void main(String[] args) {
       Scanner scan = new Scanner(System.in);
         System.out.print("Bitte geben Sie eine röm. Zahl ein: ");
         String rome = scan.nextLine();
         System.out.println("Der Wert der Zahl " + rome + " ist "
             + getValue(rome));
         scan.close();
    }

}
```